10.2.2 Theories of Catalysis

For a chemical reaction to occur, the reactants are to be activated to form the activated complex. The energy required for the reactants to reach the activated complex is called the activation energy. The activation energy can be decreased by increasing the reaction temperature. In the presence of a catalyst, the reactants are activated at reduced temperatures in otherwords, the activation energy is lowered. The catalyst adsorbs the reactants activates them by weakening the bonds and allows them to react to form the products.

As activation energy is lowered in presence of a catalyst, more molecules take part in the reaction and hence the rate of the reaction increases.

The action of catalysis in chemical reactions is explained mainly by two important theories. They are

(i) the intermediate compound formation theory

(ii) the adsorption theory.

The intermediate compound formation theory

A catalyst acts by providing a new path with low energy of activation. In homogeneous catalysed reactions a catalyst may combine with one or more reactant to form an intermediate which reacts with other reactant or decompose to give products and the catalyst is regenerated.

Consider the reactions:

$A+B \rightarrow AB$ (1)

$A+C \rightarrow AC$ (intermediate) (2)

C is the catalyst

$AC+B \rightarrow AB+C$ (3)

Activation energies for the reactions (2) and (3) are lower compared to that of (1). Hence the formation and decomposition of the intermediate accelerate the rate of the reaction.

Example 1

The mechanism of Fridel crafts reaction is given below

$$C_6H_6+CH_3Cl \xrightarrow[AlCl_3]{\text{anhydrous}} C_6H_5CH_3+HCl$$

The action of catalyst is explained as follows

$$CH_3Cl+AlCl_3 \rightarrow [CH_3]^+[AlCl_4]^-$$

It is an intermediate.

$$C_6H_6+[CH_3^+][AlCl_4]^- \rightarrow C_6H_5CH_3+AlCl_3+HCl$$

Example 2

Thermal decomposition of $KClO_3$ in presence of MnO_2 proceeds as follows.

Steps in the reaction $2KClO_3 \rightarrow 2KCl+3O_2$ can be given as

XII U10-Surface Chemistry.indd 80 11-01-2021 15:33:33